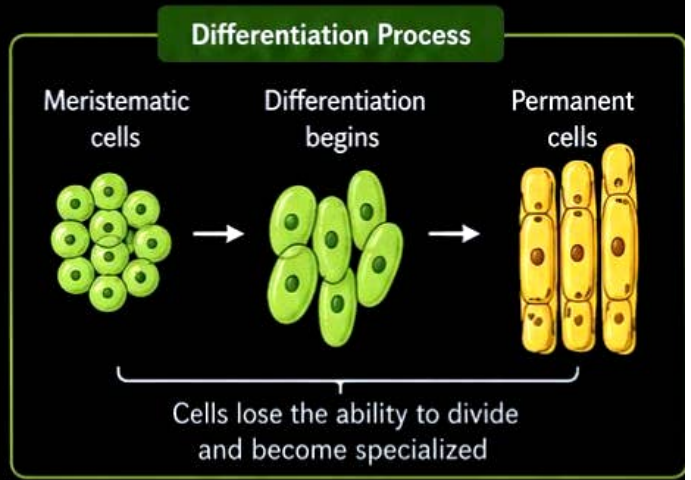


Differentiation

Differentiation = the process by which meristematic tissue becomes specialized (in structure and function) to form permanent tissue — cells lose the ability to divide and take up specific roles like support, transport, or storage.

- Some new cells stay meristematic (keep dividing)
- Other cells differentiate → become **permanent tissue**



3.2.4 Permanent Tissues

Permanent tissues = simple (one cell type) or complex (more than one cell type).

Classified by function into 3 groups:

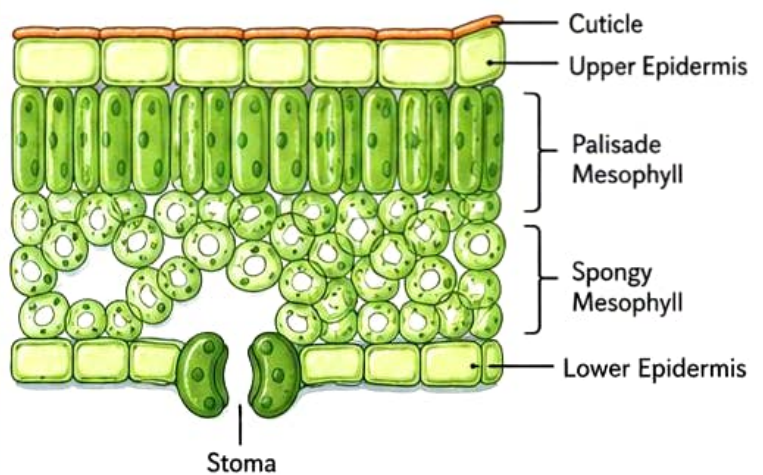
(i) Protective Tissue — Epidermis

- Forms the outermost layer of the plant body
- Single layer of flat, tightly packed, rectangular cells
- Covered by a waxy layer called **cuticle** (secreted by epidermis) — reduces water loss
- In dry-habitat plants, cuticle layer is thicker (less transpiration loss)

Functions of Epidermis:

- Protects against mechanical injury
- Prevents water loss
- Protects against microorganisms/parasites
- Protects against extreme environmental conditions

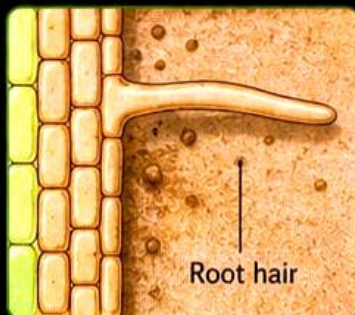
T.S. of Dicot Leaf (showing Epidermis)



Special Modifications

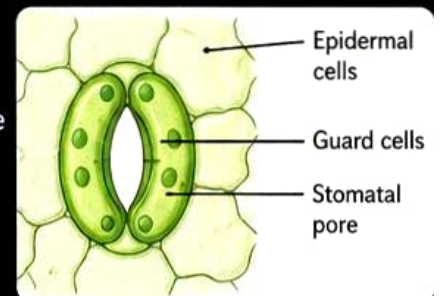
1. Root Hair

Hair-like projections from epidermal cells in roots; increase surface area for absorption of water & minerals.



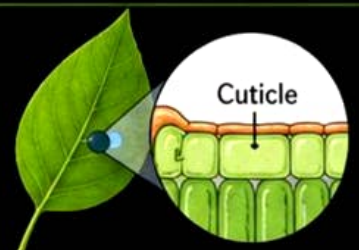
2. Stomata

Pores in leaf epidermis; allow gaseous exchange and transpiration (evaporation of water vapour).

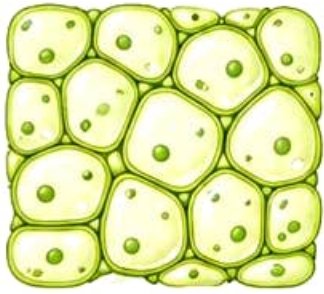


Q: What is the role of the cuticle?

A: The cuticle is a waxy layer over the epidermis that reduces water loss and protects against mechanical injury and invasion by parasites/pathogens.



PARENCHYMA

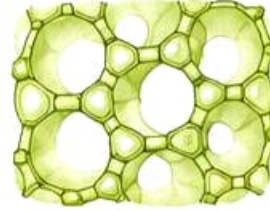


Thin walls

Living

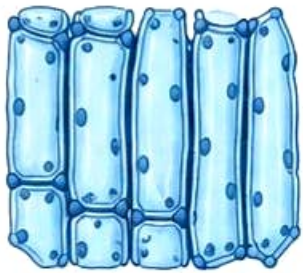


Loosely packed, intercellular spaces present



- Stores food
- Photosynthesis (in green parts)
- Forms air spaces in aquatic plants (helps floating)

COLLENCHYMA

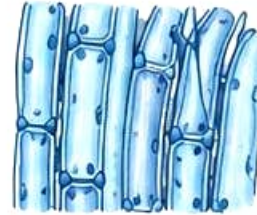


Unevenly thickened (corners thickened due to pectin)

Living

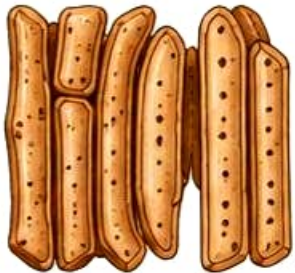


Flexible like rubber



- Gives flexibility & support
- Lets stems/tendrils bend without breaking

SCLERENCHYMA

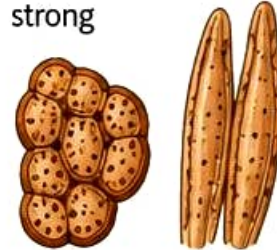


Thick walls (due to **lignin** deposition)

Mostly Dead



Hard and strong



Provides mechanical strength — found in stems, veins, seed coats, nut shells, coconut husk, walnut shell

WHERE DO WE FIND THEM?

1. Parenchyma



Mesophyll of leaf



Storage in fruits



Air spaces in aquatic plants

Key Point:

Loosely packed cells with intercellular spaces.

2. Collenchyma



In leaf stalk (petiole)



In young stems & tendrils

Key Point:

Provides support with flexibility.

3. Sclerenchyma



In stem (wood)



In leaf veins



In nut shells & seed coats

Key Point:

Provides strength and hardness.

Q

Why does a fresh twig bend but a dry twig break?

A

A fresh twig has collenchyma cells with flexible, unevenly thickened walls that provide strength with flexibility, so it bends without breaking. In a dry twig, cells lose water and the tissues become rigid and brittle (mainly due to sclerenchyma), so it breaks.

Fresh Twig



Bends

Dry Twig



Breaks



Remember:

Parenchyma (Store & photosynthesise) • Collenchyma (Support & flexibility) • Sclerenchyma (Strength)